Caleb G. Strout

Jurupa Valley, CA | caleb.g.strout@gmail.com | (909)-682-4362 | Portfolio | LinkedIn

EDUCATION

Vanderbilt University **Nashville, TN**
Bachelor of Engineering in Mechanical Engineering (Concentration in Aerospace) *May 2027*

- **Relevant Coursework:** *Aerospace Propulsion, Heat Transfer, Fluid Mechanics, Thermodynamics, Mechanical Design, Dynamics, Statics, System Dynamics, Mechanics of Materials, Machine Analysis and Design*
- **GPA:** *3.91, Dean's List (all semesters)*

SKILLS

- **Programming Languages:** Python, MATLAB, LabVIEW, R, C++
- **Tools:** CATIA**,** Creo, SOLIDWORKS CAD/CAM, OnShape, AutoCAD, Ansys, Arduino, 3-D Printing, Raspberry PI, CNC, Milling

EXPERIENCE

General Atomics **Adelanto, CA**
Fuel Systems Engineer Intern *June 2026 – Aug 2026*

- Designed testing of manifold for Mojave STOL to verify CFD analysis to ensure variety of solenoid valves are compatible
- Designed test setup of Secon Heat Exchanger for Mojave STOL on engine test stand to be conducted in October 2026
- Analyzed YFQ-42A flight data to increase pitch and roll degree range for fuel mismatch to ensure safety with leakages and CG balance
- Conducted CG analysis of varying fuel bladder volumes for Mojave STOL through creating high order polynomial functions based on loads for fuselage tanks and wing cells
- Wrote official test procedure revisions for MIL spec hot/regular refuel tests for YFQ-42A

Whisper Aero **Nashville, TN**
Test Engineer Intern *Feb 2026 – May 2026*

- Designed and assembled spin-to-burst test chamber to evaluate failure in the bladed disk of a propulsor scaled for a leaf blower product, stepping from 20,000 rpm to 35,000 rpm
- Prototyped and validated a stubby nozzle attachment through blow force, acoustic, and balancing tests to ensure more concentrated air flow as fluid exits the duct of the leaf blower, resulting in an intensive sound of 50 db(A) at 50 feet away and a maximum blow force of 30 N within product specifications

Vanderbilt Aerospace Design Laboratory **Nashville, TN**
Summer Research Intern *Jun 2025 – Aug 2025*

- Engineered gear system to deploy fins simultaneously with one servo motor, inducing different drag forces dependent on the fins' angular positions
- Integrated a VN-100 IMU with a Raspberry Pi 4 using Python to collect apogee related data to determine when fins should be deployed with 95% accuracy
- Introduced induced drag functionality to an existing rocket simulation software using MATLAB to model CD increase of rocket when fins deploy at a specific altitude, velocity, and time based on a predicted apogee with 95% accuracy
- Launched 68" subscale rocket on 4 separate occasions with target apogees of 750 feet that successfully deployed fins and recorded apogees within 20 feet of the target

PROJECTS

NASA USLI Launch Competition | *SOLIDWORKS, R&D, C++, CNC, Simulations, FEA, CFD* *Aug 2025 – Present*

- Validated mechanical design choices of vehicle and payload components including payload shearing, bolt tear out, airframe loads, and coefficient of drag to ensure safety and reliability of design using Ansys FEA and CFD products
- Created a hardware in the loop test bench to validate payload electronics and simulate rocket flight without motor usage using C++, Python, and TCP
- Machined and manufactured vehicle components including fiber glass nose cone, carbon fiber fins, and aluminum bulk plates using CNC manufacturing and machining
- Designed soil excavator payload that transfers soil to the vehicle to test conductivity, pH level, and nitrogen-nitrate content of the soil

Autonomous Lunar Robot | *SOLIDWORKS, R&D* *Aug 2024 – Present*

- Designed and machined u-channel and tension belt retainer in SOLIDWORKS to improve electromechanical functions of robot designed to dig and transport lunar dust
- Provisioned an enhanced drivetrain system of the robot in preparation to fully autonomize the robot's operations